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NET622. FA3. WEEK 7

-Design the Network topology with Packet Tracer to build a network diagram with Cisco devices, such as routers, switches and endpoints.

-Configure the IP addresses, subnet masks (automatic) and basic routing protocols to ensure the network is functional before the test.

-Document the Initial State, record the network topology, device configurations and IP addressing schemes.

-Using the ping command to check the network is working as expected.

-Introduce a Failure simulate a physical failure by removing or disconnect a cable to simulate a link failure.

-Simulate a logical failure, shut down a router or switch interface or disable a device in the Packet Tracer interface to simulate a device failure.

-Document the Failure, note the time of the failure.

-Record observations of the network behavior after the failure, such as:

-Loss of connectivity to specific devices or services.

-High latency or packet loss.

-Status changes in network monitoring tools within Packet Tracer.

-Implement and Document Recovery Actions by Logical recovery to enable the disabled interface or device and adjusting routing protocol parameters if a link was down.

-Physical recovery, reconnecting any disconnected cables and note the time it takes for the network to recover fully.

-Verify Recovery, perform connectivity tests to confirm that communication has been restored to all devices and services.